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Off-axis alignment based on optical three-dimension rendering for compact adaptive optics scanning laser ophthalmoscope

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ABSTRACT

Adaptive optics(AO) compensates for aberration between light detector and imaging target, for example, air turbulence, misaligned optics, and eye lens. This technique, initially developed to improve the performance of astronomic imaging, is leading to advances in the ophthalmic imaging field through combination with various optical imaging systems. General hardware-based AO systems need additional optics and beam paths for adjusting the beam size at the deformable mirror(DM) as well as matching the image plane. Previous research has composed the system using spherical mirrors instead of the lens to reduce aberration. Additionally, previous papers have reported off-axis type AO imaging systems for eliminating astigmatism and more compact equipment than on-axis type. This study optimized an off-axis AO-region of the AO-scanning laser ophthalmoscope(SLO) through optical design by Opticstudio[®] and three-dimensional rendering by SOLIDWORKS[®]. The rendering process, including virtual mounting, allows checking whether block or pass of the beam by optomechanics and providing relative coordinate of mounted optics. The verification method for checking the alignment of the system is the comparison between simulation and practical beam wavefront at a specific beam path.

Keywords: Adaptive optics, Scanning laser ophthalmoscope, Opticstudio, Zemax, Solidworks,

1. INTRODUCTION

Optical imaging systems could acquire distorted and blurred image data due to aberration. The aberration in the imaging system occurs from various causes including misaligned optics, eye lens, and air turbulence. Adaptive optics (AO), originally developed to enhance the performance of astronomic telescopes, reduces aberration by correcting imaging beam's distorted wavefront [1]. AO has been used for not only astronomical imaging but also ophthalmic imaging during recent decades [2-4]. After the development of the Shack-Hartmann wavefront sensor able to measure the total optical aberration of the eye, AO has complemented the performance of the retinal imaging system including scanning laser ophthalmoscope, optical coherence tomography, and fundus angiography [2-5]. Hardware type AO generally comprise the wavefront sensor and the deformable mirror for correcting distortion. These essential components need more complex beam paths and optomechanics than conventional imaging solutions without AO. Complex and extended beam paths lead to misaligned optics and expanded system size. Many researchers have composed their system by applying spherical mirrors rather than the lens to decline aberration and AO installed region. Furthermore, previous studies build the off-axis type of AO optics to reduce astigmatism, one of the crucial aberrations, and to make a narrower setup than the on-axis system [6-10]. Although the advantage of off-axis alignment, the entangled beam path as well as out of parallel with the optical table surface, cause trial and error for alignment. If the light source is invisible infrared, the blocked beam by optomechanics is not shown. To minimize trial and error, we extracted the relative coordinate of optomechanics after the three-dimensional rendering of optics and mount tools. After alignment based on relative coordinates, the assessment was performed by comparing the wavefront between optical simulation and the measured value of the actual beam.

2. METHODS AND MATERIALS

2.1 Optical simulation

The AO region of the AO-SLO system is designed in the sequential mode of Opticstudio® which provides two design options, sequential and non-sequential. The optical design comprises a collimator, a deformable mirror, two flat mirrors, two scanners, and ten spherical mirrors. The horizontal scanner is a resonant scanner (CRS 8K, Cambridge Technology, Bedford, MA, USA) and the vertical scanner is a galvo scanner (8315k, Cambridge Technology, Bedford, MA, USA). The spherical mirror is 1-inch at the anterior beam path before the scanner. The spherical mirror diameter was enlarged to 2-inch after the beam passed the scanner. The focal length of each spherical mirror is shown in table 1.

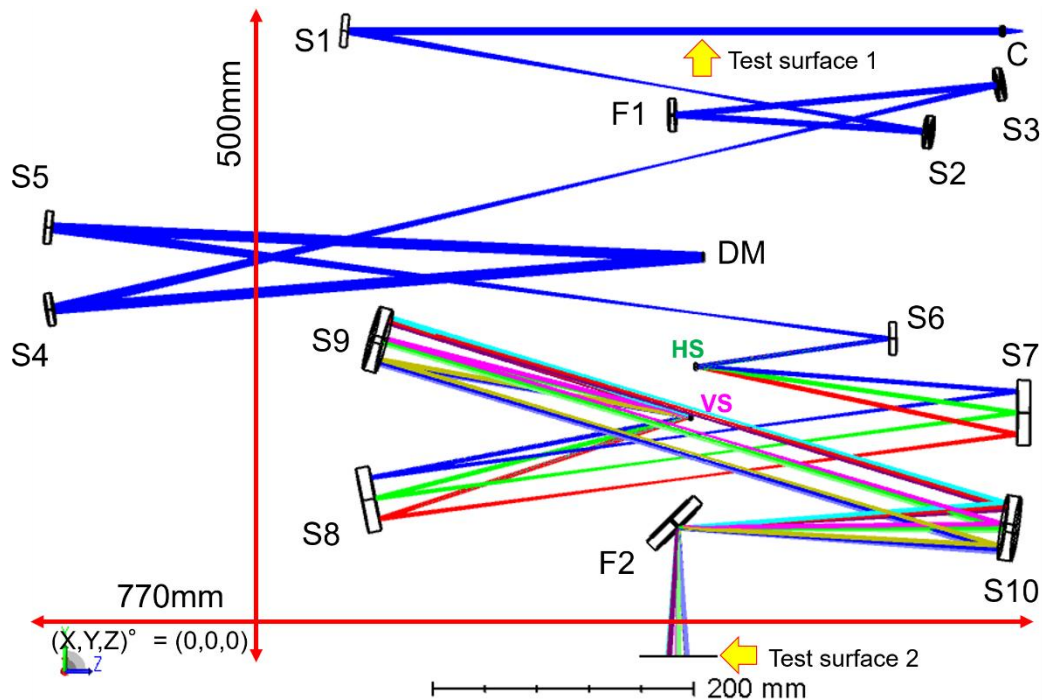


Figure 1. Top view of the optical design by Opticstudio® for the AO part of the imaging system

Table 1. Optical component list of the design.

No.	Component	Focal length	No.	Component	Focal length
1.	C ; Collimator	-	9.	S6 ; Spherical mirror 6	150mm
2.	S1 ; Spherical mirror 1	250mm	10.	HS ; Horizontal scanner (Resonant scanner)	-
3.	S2 ; Spherical mirror 2	200mm	11.	S7 ; Spherical mirror 7	250mm
4.	F1 ; Flat mirror 1	-	12.	S8 ; Spherical mirror 8	500mm
5.	S3 ; Spherical mirror 3	250mm	13.	VS ; Vertical scanner (Galvo scanner)	-
6.	S4 ; Spherical mirror 4	500mm	14.	S9 ; Spherical mirror 9	250mm
7.	DM ; Deformable mirror	-	15.	S10 ; Spherical mirror 10	250mm
8.	S5 ; Spherical mirror 5	500mm	16.	F2 ; Flat mirror 2	-

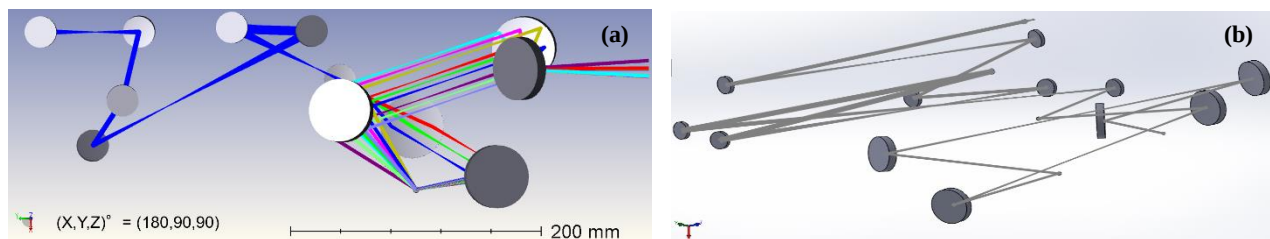


Figure 2. 3D view of the optical design. (a) side view of Fig. 1., (b) SOLIDWORKS® rendering image of the optical design extracted as CAD file

2.2 Optics and Optomechanics mounting

After the extracted optical design as a CAD file was mounted on the breadboard, rendered optomechanics, including optical posts, post holders, and clamping folks, were mounted and fixed in each location by the mating function of the SOLIDWORKS®. The function of the DimXpert location dimension provided relative coordinates of the optical posts from the temporarily mounted alignment tool at the peripheral side. The height and tilt of optics were assigned by the DimXpert function too. Actual alignment was performed using vernier calipers to measure the distance between each part.

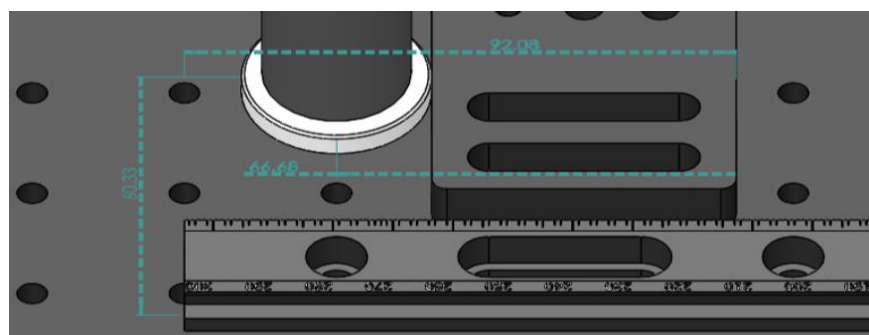


Figure 4. Location detection process by the function of DimXpert location dimension in SOLIDWORKS®

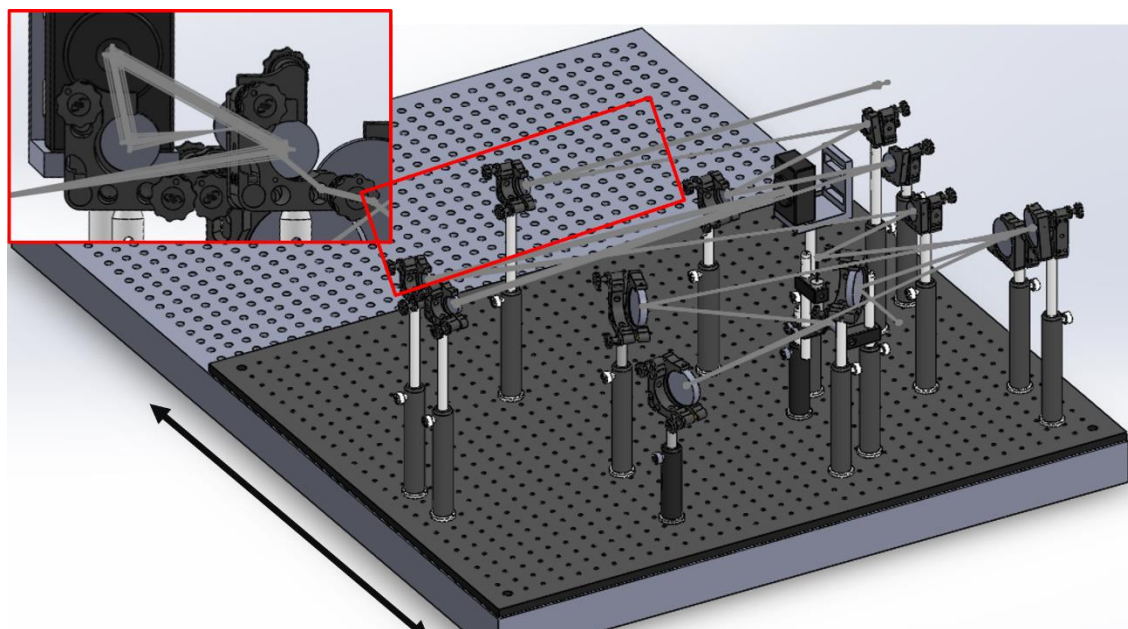


Figure 5. Assembled system image rendering by SOLIDWORKS

3. RESULTS

We aligned off-axis adaptive optics for the AO-SLO system based on optical simulation, three-dimensional rendering, and virtual mounting. All components mounted on the breadboard can move on the optical table to extend space for the remaining AO-SLO parts including the scanner controller, power system, and data acquisition. The remaining space of the optical table is shown in Fig. 5.

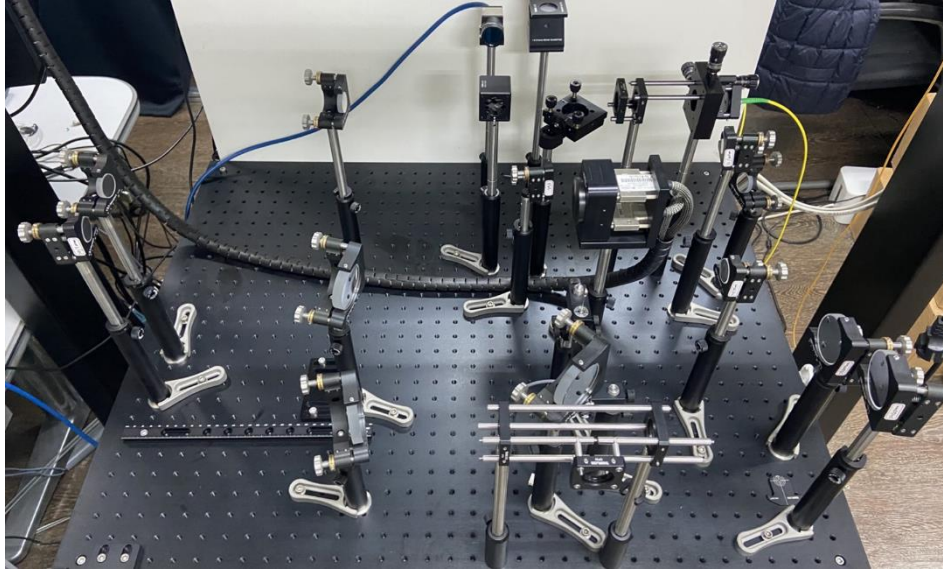


Figure 6. Image of the aligned optics for AO-SLO. The white background screen is applied to improve visibility

Alignment result verification is a wavefront comparison between simulation and actual measurement at each test surface shown in Fig.1 where the surfaces are able to measure input and output beam. We mainly compared defocus aberration, the highest aberration value in simulation. The measured defocus values in each test surface are similar to the simulated defocus value. Although a little different aberration value occurs, it could be corrected by a deformable mirror through a feedback loop correction algorithm.

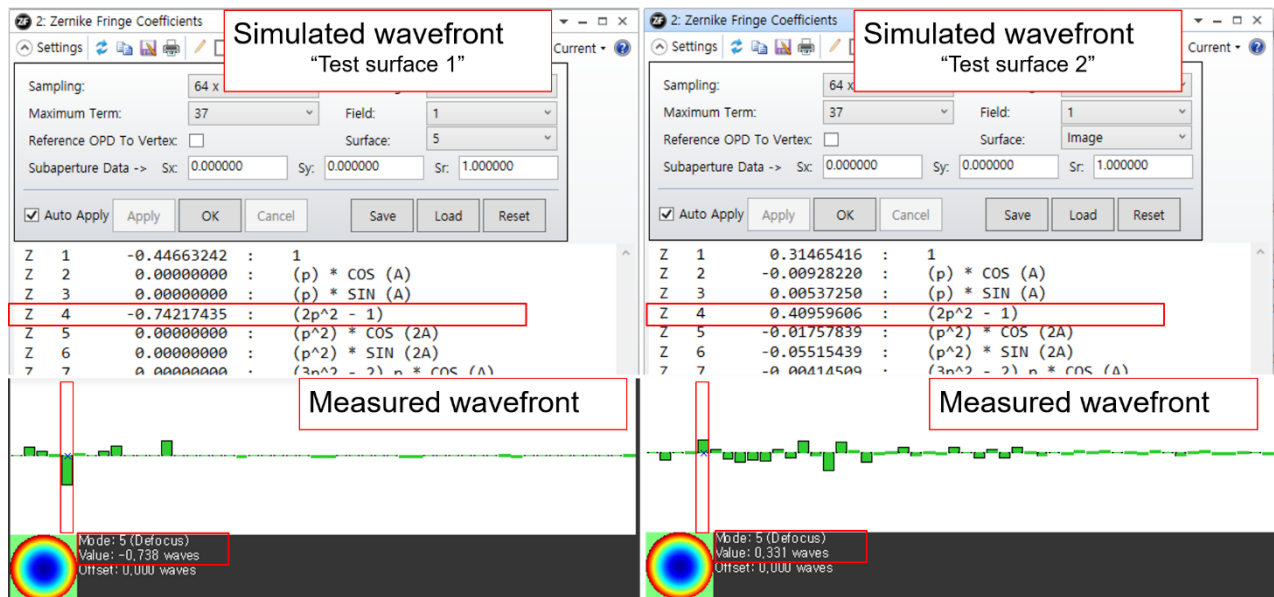


Figure 7. Comparison between simulated and measured defocus value

4. CONCLUSION

In this study, we aligned aberration correcting optics based on optical 3D rendering and simulation. In the 3D rendering software SOLIDWORKS, commercial optomechanics models which are well-mounted with the optical design, give relative coordinate for fine location. Alignment based the relative coordinate is performed with vernier calipers. Demonstration of alignment is comparing wavefront between simulation and actual setup. The simulation showed aberration via Zernike polynomials at the two test surfaces (input and output beam). Due to the defocus aberration being major in the simulation, we focus on defocus comparison for assessment. Through this study, we confirmed the feasibility of an optical imaging system setup based on 3D rendering and optical simulation. Virtual alignment based on optical design and 3D rendering helps set complex off-axis imaging systems by hand in the laboratory.

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REFERENCES

- [1] J. M. Beckers, "Adaptive Optics for Astronomy: Principles, Performance, and Applications," *Annual Review of Astronomy and Astrophysics*, 31(1), 13-62 (1993).
- [2] E. Arthur, A. E. Elsner, K. A. Sapoznik *et al.*, "Distances from capillaries to arterioles or venules measured using OCTA and AOSLO," *Investigative ophthalmology & visual science*, 60(6), 1833-1844 (2019).
- [3] D. Miller, O. Kocaoglu, Q. Wang *et al.*, "Adaptive optics and the eye (super resolution OCT)," *Eye*, 25(3), 321-330 (2011).
- [4] M. Pircher, and R. J. Zawadzki, "Review of adaptive optics OCT (AO-OCT): principles and applications for retinal imaging," *Biomedical optics express*, 8(5), 2536-2562 (2017).
- [5] T. O. Salmon, L. N. Thibos, and A. Bradley, "Comparison of the eye's wave-front aberration measured psychophysically and with the Shack-Hartmann wave-front sensor," *JOSA A*, 15(9), 2457-2465 (1998).
- [6] A. Gómez-Vieyra, A. Dubra, D. Malacara-Hernández *et al.*, "First-order design of off-axis reflective ophthalmic adaptive optics systems using afocal telescopes," *Optics Express*, 17(21), 18906-18919 (2009).
- [7] Z. Liu, O. P. Kocaoglu, R. S. Jonnal *et al.*, "Performance of an off-axis ophthalmic adaptive optics system with toroidal mirrors," *OSA Technical Digest (CD)*. AMA4.
- [8] Z. Liu, O. P. Kocaoglu, and D. T. Miller, "In-the-plane design of an off-axis ophthalmic adaptive optics system using toroidal mirrors," *Biomedical Optics Express*, 4(12), 3007-3030 (2013).
- [9] S.-H. Lee, J. S. Werner, and R. J. Zawadzki, "Improved visualization of outer retinal morphology with aberration cancelling reflective optical design for adaptive optics - optical coherence tomography," *Biomedical Optics Express*, 4(11), 2508-2517 (2013).
- [10] R. J. Zawadzki, P. Zhang, A. Zam *et al.*, "Adaptive-optics SLO imaging combined with widefield OCT and SLO enables precise 3D localization of fluorescent cells in the mouse retina," *Biomedical Optics Express*, 6(6), 2191-2210 (2015).